

DEVOPS & CLOUD ENGINEERING SERIES

The Complete DevOps Engineer Interview Guide — Exhaustive Edition

2026 Edition — Every Question Adds Value to Your Interview Preparation

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303

INTERVIEW QUESTIONS

27

TOPIC MODULES

1st

PERSON ANSWERS

2026

EDITION

What's inside:

- 1. DevOps Interview Q&A
- 2. General Networking Questions for Dev...
- 3. Subnet-Related Questions
- 4. AWS Networking Questions
- 5. Azure Networking Questions
- 6. Version Control: Git & GitHub
- 7. GitHub Actions
- 8. ArgoCD
- 9. CI/CD Pipeline
- 10. Containerization: Docker & Kuberne...
- 11. Kubernetes
- 12. Kubernetes Architecture
- 13. Ingress Controller
- 14. Infrastructure as Code: Terraform &...
- 15. Cloud Computing: AWS
- 16. Cloud Computing & Azure DevOps
- 17. Monitoring & Logging
- 18. Security in DevOps
- 19. Jenkins
- 20. Linux for DevOps
- 21. SonarQube
- 22. Trivy
- 23. Selenium
- 24. Nexus
- 25. Combined: GitHub Actions, ArgoCD &...
- 26. GitLab
- 27. Python for DevOps

Compiled and elaborated for interview preparation — practical, first-person answers.

Table of Contents

1. DevOps Interview Q&A;	Q1–Q5
2. General Networking Questions for DevOps	Q6–Q54
3. Subnet-Related Questions	Q55–Q70
4. AWS Networking Questions	Q71–Q85
5. Azure Networking Questions	Q86–Q105
6. Version Control: Git & GitHub	Q106–Q110
7. GitHub Actions	Q111–Q114
8. ArgoCD	Q115–Q118
9. CI/CD Pipeline	Q119–Q123
10. Containerization: Docker & Kubernetes	Q124–Q129
11. Kubernetes	Q130–Q134
12. Kubernetes Architecture	Q135–Q148
13. Ingress Controller	Q149–Q158
14. Infrastructure as Code: Terraform & Ansible	Q159–Q163
15. Cloud Computing: AWS	Q164–Q168
16. Cloud Computing & Azure DevOps	Q169–Q193
17. Monitoring & Logging	Q194–Q198
18. Security in DevOps	Q199–Q200
19. Jenkins	Q201–Q215

20. Linux for DevOps	Q216–Q240
21. SonarQube	Q241–Q253
22. Trivy	Q254–Q263
23. Selenium	Q264–Q273
24. Nexus	Q274–Q280
25. Combined: GitHub Actions, ArgoCD & Kubernetes	Q281–Q283
26. GitLab	Q284–Q293
27. Python for DevOps	Q294–Q303

Each module below groups related questions the way an interviewer typically sequences them — fundamentals first, then tooling, then advanced and scenario-based topics.

01

DEVOPS INTERVIEW Q&A;

Q1

What is DevOps, and why is it important?

DevOps is a set of practices that bridges development and operations by automating and integrating the processes between them, so collaboration improves and software ships faster without sacrificing reliability. It matters because it shortens development cycles, improves release efficiency through CI/CD and monitoring, and builds a culture where teams own quality end to end instead of throwing code over a wall.

Q2

Can you explain the differences between Agile and DevOps?

Agile is focused on the development process itself -- breaking work into sprints and delivering working software iteratively, with feedback coming mainly from stakeholders after each sprint. DevOps extends that same iterative philosophy across the entire lifecycle, including deployment and operations, requiring collaboration between dev, ops, and QA, and leaning heavily on automation for build, test, and deployment rather than the more manual testing Agile alone implies. DevOps also closes the feedback loop continuously from production itself -- performance monitoring, real user behavior -- not just from a sprint review.

KEY POINTS TO MENTION

- Agile: iterative development, sprint-based, stakeholder feedback
- DevOps: full lifecycle, dev+ops+QA collaboration, heavy automation, continuous production feedback

Q3

What are the key principles of DevOps?

Automation of testing, integration, and deployment to speed delivery and cut errors; close collaboration between development, QA, and operations; CI/CD so every change is automatically tested and deployed; continuous monitoring and feedback so issues surface early; infrastructure as code for consistent, versioned environments; and a culture of continuous improvement built on retrospectives and experimentation.

KEY POINTS TO MENTION

- Automation
- Collaboration
- CI/CD
- Monitoring & Feedback
- Infrastructure as Code
- Culture of improvement

Q4**How do Continuous Integration (CI) and Continuous Deployment (CD) work together?**

CI is about integrating code changes into a shared repository multiple times a day, with each integration verified through automated builds and tests so errors are caught as early as possible. CD extends that by automatically deploying the tested, integrated code to production, so any change that passes the test suite reaches users with minimal manual intervention. Together, CI keeps the codebase stable through frequent verification, while CD makes sure that stable code actually reaches production quickly and reliably.

Q5**What challenges have you faced implementing DevOps in previous projects?**

Cultural resistance is usually the biggest one -- dev and ops teams that are used to working in silos don't automatically start collaborating just because you introduce a pipeline. Tool integration is another, especially with legacy systems that weren't built with automation in mind; skill gaps on tools like Jenkins, Docker, or Kubernetes can slow early adoption; managing infrastructure through IaC requires a mindset shift for teams used to manual provisioning; and folding security checks into CI/CD, real DevSecOps, adds real complexity when deployments are frequent and compliance still needs to be satisfied.

02

GENERAL NETWORKING QUESTIONS FOR DEVOPS

Q6

What is the OSI Model?

It's the 7-layer framework I use as a mental checklist for troubleshooting -- physical for raw hardware transmission, data link for framing and error detection, network for IP routing, transport for reliable end-to-end delivery, session for managing application sessions, presentation for encryption and format translation, and application for the actual services users interact with. When something's broken I work up that stack instead of guessing where the problem is.

KEY POINTS TO MENTION

- Physical, Data Link, Network, Transport, Session, Presentation, Application

Q7

What is TCP/IP?

TCP/IP is the practical 4-layer protocol suite -- link, internet, transport, application -- that actually implements reliable communication across real networks, and it's what the OSI model's more theoretical 7 layers map onto in practice.

Q8

What is DNS, and why is it important?

DNS resolves human-readable domain names into IP addresses, which is what makes the internet usable without everyone memorizing IPs. In DevOps it matters a lot beyond just browsing -- service discovery, load balancer failover, and CDN routing all depend on DNS behaving correctly and propagating in a predictable time.

Q9

What is a firewall?

A firewall controls inbound and outbound traffic based on defined security rules, protecting systems from unauthorized access. In cloud environments that concept shows up as security groups and network ACLs, which I treat with the same seriousness as a traditional on-prem firewall.

Q10

What is NAT (Network Address Translation)?

NAT translates private IP addresses to a public one so devices on a private network can reach the internet without each one needing its own public IP. In AWS that's exactly what a NAT Gateway does for instances in a private subnet.

Q11

Explain the difference between TCP and UDP.

TCP is connection-oriented and reliable -- it guarantees delivery and ordering through acknowledgments and retransmission, which is why I use it for anything where correctness matters more than speed, like API calls or database connections. UDP is connectionless and faster but doesn't guarantee delivery, so it fits use cases like DNS queries or metrics where an occasional dropped packet is an acceptable trade-off for lower latency.

Q12**What is a VPN, and why is it used in DevOps?**

A VPN creates a secure, encrypted connection over the internet, and in DevOps I rely on it for secure remote access to private infrastructure -- reaching internal servers or a private VPC without exposing them directly to the public internet.

Q13**What is Load Balancing?**

Load balancing distributes incoming traffic across multiple servers so no single instance is overwhelmed, and it's also what gives me automatic health-check-based failover -- an unhealthy instance simply stops receiving traffic without anyone manually intervening.

Q14**What is a Proxy Server?**

A proxy server sits between a client and the internet, acting as an intermediary that can improve security by hiding the client's real address and improve performance through caching. I've used forward proxies for controlling outbound access from a private environment and reverse proxies in front of application servers.

Q15**What is a Subnet Mask?**

A subnet mask defines which portion of an IP address is the network and which portion is the host, letting a large network be segmented into smaller, more manageable pieces -- something like 255.255.255.0 marks the first 24 bits as network and the rest as host.

Q16**What is Round-Robin DNS and how does it benefit DevOps?**

Round-robin DNS returns multiple IP addresses for the same domain name in rotating order, giving a simple form of load distribution across servers. It's a lightweight way to add resilience and spread load without needing a dedicated load balancer, though it lacks real health-check awareness compared to an actual load balancer.

Q17**How do Firewall Rules apply to DevOps?**

Firewall rules -- security groups and NACLs in cloud terms -- restrict which traffic can reach a CI/CD environment or production application. I use them to limit exposure, allowing only the specific ports and sources a service actually needs, especially tightening things down for production compared to more permissive dev environments.

Q18**What is a Packet Sniffer and its role in DevOps?**

A packet sniffer like Wireshark or tcpdump captures and inspects raw network traffic, which I've used to troubleshoot connectivity issues, debug microservice communication, or figure out exactly where a pipeline-related network call is failing when higher-level logs aren't giving enough detail.

Q19 How does IPsec VPN assist DevOps?

IPsec VPNs create secure, encrypted site-to-site or client connections, which is what lets remote DevOps engineers or on-prem systems securely reach private cloud infrastructure without that traffic traversing the public internet unencrypted.

Q20 What is the difference between Routing and Switching in DevOps?

Routing manages traffic between different networks -- important for multi-cloud or hybrid setups where traffic crosses network boundaries. Switching handles traffic within a single network or data center, which matters for efficient intra-cluster or intra-VPC communication.

Q21 Why is Network Topology important in DevOps?

Understanding topology -- how nodes and network paths are physically or logically arranged -- helps me design infrastructure that's resilient and scalable, and helps me reason about traffic flow when diagnosing latency or connectivity issues inside a cluster.

Q22 How does the TCP 3-Way Handshake apply to DevOps?

SYN, SYN-ACK, ACK establishes a connection before any data flows, and understanding it is genuinely useful for troubleshooting -- if a handshake never completes, I know it's a network or firewall-level block, versus a connection that completes but the application itself hangs, which points to an application-layer problem instead.

Q23 What are CIDR Blocks and how do they assist in DevOps?

CIDR blocks define network ranges in the IP/prefix-length format, and I use them constantly for efficient, flexible network segmentation in cloud VPC design -- separating dev, test, and production into distinct ranges while making efficient use of the available address space.

Q24 How is Quality of Service (QoS) utilized in DevOps?

QoS prioritizes certain types of network traffic over others, which matters when I need to make sure a latency-sensitive or resource-intensive service gets sufficient bandwidth even when the network is under contention from less critical traffic.

Q25 What role do Network Switches play in DevOps?

Switches manage local traffic within a private network or data center, forwarding frames based on MAC address -- essential for efficient on-premise service communication in hybrid DevOps environments where not everything has moved to the cloud.

Q26**How are Broadcast Domains relevant to DevOps?**

A broadcast domain is the set of devices that receive broadcast traffic from each other, and keeping these appropriately scoped in network design minimizes unnecessary traffic and avoids performance issues that come from an overly large flat network.

Q27**What is Tunneling and how is it used in DevOps?**

Tunneling encapsulates one network protocol inside another to create a secure or otherwise unsupported path between networks -- SSH tunnels and VPNs are common examples I use for securely reaching cloud resources or bridging separate network environments.

Q28**How is EIGRP used in DevOps?**

EIGRP is a Cisco routing protocol I've mostly encountered in legacy, on-prem environments for managing internal routing efficiently -- it's less relevant in pure cloud-native setups but still shows up in hybrid infrastructure with a traditional networking footprint.

Q29**What is the role of DNS A and CNAME Records in DevOps?**

An A record maps a domain directly to an IP address, while a CNAME record maps a domain to another domain name. I use A records for root domains pointing at a fixed IP and CNAMEs for subdomains pointing at services like a load balancer or CDN endpoint whose underlying IP might change.

Q30**How do Latency and Throughput impact DevOps?**

Latency and throughput directly shape user-perceived performance, especially in distributed systems where a single user request might hop across several services. I monitor both because a service can have plenty of throughput capacity while still delivering a poor experience due to high latency on the critical path.

Q31**Why is DNS Propagation important for DevOps?**

DNS changes don't take effect everywhere instantly -- propagation delay depends on TTLs and resolver caching. I plan DNS cutovers around that, lowering TTLs in advance of a planned change so the transition happens faster and doesn't cause a prolonged period where some users hit the old endpoint.

Q32**How does ARP Poisoning affect DevOps?**

ARP poisoning is an attack where a malicious actor sends forged ARP messages to associate their MAC address with another device's IP, letting them intercept traffic. It's a network security risk that DevOps teams need to defend against with proper network segmentation and monitoring, particularly in shared or less-trusted network segments.

Q33**What is a Route Table and how is it used in DevOps?**

A route table controls how traffic flows between subnets and out to gateways -- in AWS it's the actual mechanism that determines whether a subnet is public or private, based on whether its route table sends 0.0.0.0/0 traffic to an internet gateway or a NAT gateway.

Q34**How does Mesh Topology benefit DevOps?**

A mesh topology, where nodes connect to multiple other nodes rather than a single central point, gives redundancy and failover -- which is exactly the resilience model behind Kubernetes cluster networking and service mesh architectures like Istio, where losing one path doesn't isolate a service.

Q35**How does DNS Failover support DevOps?**

DNS failover uses health checks to automatically redirect traffic away from an unhealthy endpoint to a backup, which is how I've implemented cross-region failover in Route 53 -- if the primary region's health check fails, traffic automatically shifts to the secondary without manual DNS changes.

Q36**What is an Access Control List (ACL) in DevOps?**

An ACL restricts access to specific resources based on rules, and I commonly define these as part of infrastructure-as-code configurations -- like a Network ACL in a VPC -- to enforce secure access at the subnet level as code rather than a manually managed console setting.

Q37**What is a Point-to-Point Connection in DevOps?**

A point-to-point connection is a direct link between two networks, commonly used in hybrid environments to securely connect on-prem infrastructure to a cloud VPC -- AWS Direct Connect is a real example of exactly this kind of dedicated link.

Q38**How does Split-Horizon work in DevOps?**

Split-horizon DNS resolves the same domain name differently depending on whether the query comes from inside or outside the network, which I've used to let internal services reach each other via internal IPs while external users are routed through a public-facing endpoint for the same hostname.

Q39**What is Packet Filtering in DevOps?**

Packet filtering inspects individual packets against a rule set and allows or blocks them accordingly -- it's the fundamental mechanism behind both traditional firewalls and cloud security groups, enforcing which traffic is allowed to reach an application.

Q40**How do VPN Tunnels aid DevOps?**

VPN tunnels secure the connection between on-prem infrastructure and cloud environments, which is essential in hybrid cloud setups where sensitive traffic needs to stay encrypted and private as it crosses between two otherwise separate network environments.

Q41**How are DNS MX Records used in DevOps?**

MX records specify the mail servers responsible for receiving email for a domain -- as a DevOps engineer, I make sure these are configured correctly whenever we're setting up email services or notification systems that depend on outbound or inbound mail actually routing to the right place.

Q42**What is Routing Convergence and its importance in DevOps?**

Convergence is the time it takes for routers to synchronize their routing tables after a network change. Fast convergence matters for minimizing downtime during failover events in cloud networking, since a slow-converging network means longer periods where traffic might be dropped or misrouted after a change.

Q43**What is a DHCP Scope and how does it help DevOps?**

A DHCP scope is the range of IP addresses a DHCP server can automatically assign, which simplifies IP management in on-prem or private cloud environments by automating address allocation instead of manually tracking which IPs are in use.

Q44**How do Symmetric and Asymmetric Encryption support DevOps?**

Symmetric encryption uses one shared key and is fast, which is why it's used for bulk data encryption once a connection is established. Asymmetric encryption uses a public/private key pair and is slower but solves the key-exchange problem securely -- both work together in protocols like TLS and SSH, which I rely on constantly for securing data in transit.

Q45**How does Network Latency affect DevOps?**

Low latency matters a lot for real-time or interactive applications, and I use monitoring tools to track latency trends specifically so I can catch and troubleshoot degradation in a pipeline or service before it becomes a user-visible problem.

Q46**What is the role of a Hub in DevOps?**

A hub is a basic networking device that broadcasts everything it receives to all connected ports -- I've only really seen these in small test environments or legacy office networks, since switches are strictly better for anything beyond the simplest setup.

Q47**How does Open Shortest Path First (OSPF) contribute to DevOps?**

OSPF is a dynamic routing protocol that lets routers automatically adjust routing tables in response to network changes, giving fault tolerance and efficient path selection -- important for DevOps teams managing resilient internal network routing in hybrid or on-prem-heavy environments.

Q48**How does a DMZ (Demilitarized Zone) apply in DevOps?**

A DMZ isolates public-facing services in their own network segment, acting as a buffer between the internet and internal, more sensitive networks. I've used this pattern for production environments where the public-facing load balancer or API gateway sits in the DMZ while backend services stay fully isolated from direct internet exposure.

Q49**What is a Service Level Agreement (SLA) in DevOps?**

An SLA defines the uptime and performance commitments made to customers, often with contractual consequences if breached. As a DevOps engineer I monitor the underlying metrics closely enough to know before the SLA is at risk, not just after it's already been breached.

Q50**What are Sticky Sessions and how are they used in DevOps?**

Sticky sessions configure a load balancer to consistently route a given user's requests to the same backend instance, which matters for stateful applications that store session data locally rather than in a shared external store. I generally prefer designing stateless services that don't need sticky sessions at all, since they scale and fail over more cleanly.

Q51**How does a Subnet Mask work in DevOps?**

Subnetting lets me segment a network to isolate environments -- dev, test, prod -- from each other, which improves both security, by limiting blast radius, and traffic management, by keeping broadcast domains appropriately scoped.

Q52**How is Multicast used in DevOps?**

Multicast efficiently delivers the same data to multiple receivers simultaneously without duplicating traffic for each one, which is useful in environments like Kubernetes clusters where certain real-time state updates need to reach many nodes at once.

Q53**What is Port Mirroring and how does it help DevOps?**

Port mirroring copies network traffic from one port to another for monitoring purposes without disrupting the original traffic flow -- I've used this for deep troubleshooting of microservice communication issues where application-level logs alone weren't giving enough visibility into what was actually happening on the wire.

Q54

How does Zero Trust Architecture relate to DevOps?

Zero Trust assumes nobody, inside or outside the network, is trusted by default -- every request has to prove its identity and authorization on its own merits. I implement this in DevOps through things like mutual TLS between services and scoped IAM roles per workload, rather than relying on network location as an implicit trust boundary.

03

SUBNET-RELATED QUESTIONS

Q55

What is Subnetting?

Subnetting is dividing a larger network into smaller, more manageable sub-networks, which gives better IP address management, improved performance by shrinking broadcast domains, and stronger security by isolating network segments from each other.

Q56

Why is Subnetting important in DevOps?

It lets me isolate environments -- dev, test, production -- from each other, allocate IP address space efficiently instead of wasting it, and reduce unnecessary broadcast traffic, all of which directly supports both security and performance in the environments I manage.

Q57

What is a Subnet Mask?

A subnet mask is a 32-bit number that splits an IP address into its network and host portions -- 255.255.255.0 is a common example, meaning the first 24 bits identify the network and the remaining 8 identify the individual host.

Q58

What is CIDR (Classless Inter-Domain Routing)?

CIDR replaced the old class-based (A/B/C) addressing system with a flexible notation combining the IP address and prefix length, like 192.168.1.0/24, which lets networks be sized exactly to what's needed instead of forced into fixed class boundaries.

Q59

What is the difference between Public and Private IP Subnets?

Public IP subnets are assigned to resources that need to be reachable from the internet, like a web-facing load balancer. Private IP subnets are for internal resources that shouldn't be directly internet-accessible, like application or database tiers -- that public/private distinction is exactly how I structure VPC subnets in Terraform.

Q60

How do you calculate the number of subnets and hosts in a given subnet?

Number of subnets is 2 to the power of the bits borrowed from the host portion, and hosts per subnet is 2 to the power of the remaining host bits, minus 2 to account for the network and broadcast addresses. I use this constantly when deciding how to size VPC subnets so I don't over- or under-allocate address space.

KEY POINTS TO MENTION

- Subnets = 2^n (n = borrowed bits)
- Hosts per subnet = $2^h - 2$

Q61**Given 192.168.1.0/24, if you borrow 2 bits for subnetting, what do you get?**

Borrowing 2 bits gives a new mask of 255.255.255.192, or /26, which creates 4 subnets of 62 usable hosts each -- 2 squared for subnets, 2 to the 6th minus 2 for hosts, since 6 bits remain for hosts.

KEY POINTS TO MENTION

- 4 subnets, 62 usable hosts each

Q62**What is the difference between subnet mask 255.255.255.0 and 255.255.255.128?**

255.255.255.0, a /24, allows 256 addresses with 254 usable hosts and is typical for a moderately sized network. 255.255.255.128, a /25, splits that same original /24 space into two subnets of 128 addresses each, 126 usable hosts, which I'd use when I want two smaller isolated segments instead of one larger flat one.

Q63**How do you subnet 192.168.1.0/24 into 4 equal subnets?**

Borrowing 2 bits gives a /26 mask, 255.255.255.192, producing four subnets: 192.168.1.0/26, 192.168.1.64/26, 192.168.1.128/26, and 192.168.1.192/26 -- each with 62 usable hosts.

Q64**What are the valid IP address ranges for a 192.168.0.0/28 network?**

A /28 gives 16 total addresses: network address 192.168.0.0, usable range from 192.168.0.1 through 192.168.0.14, and broadcast address 192.168.0.15 -- 14 usable host addresses in total.

Q65**What is VLSM (Variable Length Subnet Mask) and when do you use it in DevOps?**

VLSM allows different subnet sizes within the same network rather than forcing every subnet to be identical, which is exactly what I need in a real VPC design -- a database tier might only need a handful of IPs while an application tier needs room to scale, and VLSM lets me size each appropriately instead of wasting address space.

Q66**What is the difference between a /24 and a /30 subnet?**

A /24, mask 255.255.255.0, gives 254 usable hosts, suitable for a general-purpose subnet. A /30, mask 255.255.255.252, gives just 2 usable hosts, which is exactly the size I'd use for a point-to-point link where only two devices ever need to talk to each other directly.

Q67**How do you handle subnetting in a Kubernetes environment?**

Kubernetes needs CIDR ranges allocated for nodes, pods, and services, and getting that sizing right up front matters a lot because it's painful to change after the cluster is running -- I make sure the pod CIDR is large enough to support the cluster's eventual scale, since running out of pod IPs is a hard wall to hit later.

Q68**What are Supernet, and how are they different from Subnets?**

A supernet combines several smaller networks into one larger one by reducing the subnet mask size -- the opposite operation from subnetting. It's useful for reducing the number of entries in a routing table when you have many contiguous smaller networks that can be summarized as one route.

Q69**What is a Subnetting Table, and how is it useful in DevOps?**

A subnetting table lays out different subnet sizes against the number of hosts and subnets each produces, which I use as a quick reference when planning a new VPC's address space so I don't have to recalculate the math from scratch every time.

Q70**How does CIDR notation improve IP address management in DevOps?**

CIDR gives flexible, efficient address allocation instead of the rigid class-based system, letting me size a subnet to exactly what an environment needs -- a small management subnet doesn't have to waste an entire class C's worth of addresses the way older addressing schemes would have forced.

04

AWS NETWORKING QUESTIONS

Q71

What is a VPC in AWS?

A VPC is a private, logically isolated network within AWS where I control the IP address range, subnets, route tables, and gateways -- it's the foundational building block for every piece of AWS infrastructure I provision.

Q72

How do Security Groups work in AWS?

Security Groups are stateful, instance-level virtual firewalls -- I define allow rules for inbound and outbound traffic, and because they're stateful, a response to an allowed inbound request is automatically permitted outbound without needing a matching explicit rule.

Q73

What is an Internet Gateway in AWS?

An Internet Gateway attaches to a VPC and enables two-way internet connectivity for resources in public subnets, which is the piece that actually makes a subnet "public" once paired with the right route table entry.

Q74

What is a NAT Gateway?

A NAT Gateway lives in a public subnet and lets instances in private subnets initiate outbound internet connections -- for updates, external API calls -- without being directly reachable from the internet themselves.

Q75

What is Route 53?

Route 53 is AWS's DNS service, and I use it for more than basic hosting -- health-check-based failover routing and weighted routing policies let it actively contribute to application availability, not just resolve names.

Q76

What is an Elastic Load Balancer (ELB)?

ELB distributes incoming traffic across multiple instances or targets, giving both scalability and fault tolerance -- if one target becomes unhealthy, the load balancer stops routing to it automatically based on configured health checks.

Q77

What is AWS PrivateLink?

PrivateLink provides private connectivity between VPCs and AWS services without that traffic ever traversing the public internet -- I've used it to expose an internal service to another team's VPC securely, without a full VPC peering connection or public exposure.

Q78 What is a Transit Gateway?

Transit Gateway acts as a central hub connecting multiple VPCs and on-prem networks, which dramatically simplifies networking once you have more than a couple of VPCs that need to talk to each other, replacing what would otherwise be a messy mesh of individual VPC peering connections.

Q79 What are Subnets in AWS?

Subnets are segments within a VPC's IP range used to organize resources and control traffic flow -- whether a subnet is public or private comes down entirely to its associated route table, not any inherent property of the subnet itself.

Q80 What is AWS Direct Connect?

Direct Connect provides a dedicated, private, low-latency physical connection between on-prem data centers and AWS, bypassing the public internet entirely -- I'd recommend it for consistent high-bandwidth needs or strict compliance requirements around data never touching the public internet.

Q81 What is VPC Peering?

VPC Peering creates a direct network connection between two VPCs so resources in each can communicate as if they were on the same network. It's simple to set up but doesn't scale well past a handful of VPCs since it isn't transitive -- each pair needs its own dedicated peering connection.

Q82 What is an Egress-Only Internet Gateway?

It's specifically for IPv6 traffic -- it allows outbound connectivity from a VPC while blocking any unsolicited inbound traffic, functioning as the IPv6 equivalent of what a NAT Gateway does for IPv4.

Q83 Difference between Security Groups and Network ACLs?

Security Groups are stateful and operate at the instance level -- return traffic is automatically allowed. Network ACLs are stateless and operate at the subnet level -- I have to explicitly allow both inbound and outbound traffic. I use security groups for day-to-day access control and NACLs for broader subnet-level rules like explicitly blocking a known bad IP range.

Q84 What is AWS Global Accelerator?

Global Accelerator routes traffic through AWS's global backbone network instead of the public internet's variable path, reducing latency and improving performance and availability for globally distributed users hitting an application.

Q85**How do you monitor network traffic in AWS?**

VPC Flow Logs capture accepted and rejected traffic at the ENI level, which is my first stop for understanding whether traffic is being blocked and where, and I pair that with CloudWatch metrics and alarms for ongoing visibility into network-level health.

05

AZURE NETWORKING QUESTIONS

Q86

What is a VNet (Virtual Network) in Azure?

A VNet is Azure's equivalent of a VPC -- a private, isolated network where I control IP address space, subnets, and routing to securely connect and manage resources.

Q87

What are Network Security Groups (NSGs) in Azure?

NSGs filter inbound and outbound traffic to Azure resources at the subnet or NIC level, acting as Azure's version of a virtual firewall -- functionally similar to AWS security groups, just applied slightly differently depending on whether you attach them to a subnet or an individual network interface.

Q88

What is an Application Gateway in Azure?

Application Gateway is a Layer 7 load balancer with built-in Web Application Firewall protection, handling HTTP/HTTPS routing decisions based on URL path or host, similar in role to an AWS ALB with WAF attached.

Q89

How does Azure Load Balancer work?

Azure Load Balancer distributes traffic among VMs at Layer 4 to improve availability and reliability, which I'd use for TCP/UDP-level load distribution rather than the HTTP-aware routing an Application Gateway provides.

Q90

What is Azure Traffic Manager?

Traffic Manager is a DNS-based routing service that directs traffic across Azure regions globally, similar in purpose to Route 53's latency-based or geo-routing policies in AWS.

Q91

What is a VPN Gateway in Azure?

A VPN Gateway creates a secure, encrypted connection between an Azure VNet and an on-prem network, which is the go-to option for hybrid connectivity when a dedicated ExpressRoute circuit isn't justified by the traffic volume or budget.

Q92

What is Azure ExpressRoute?

ExpressRoute provides a private, dedicated, high-bandwidth connection between on-prem infrastructure and Azure that bypasses the public internet -- Azure's direct equivalent to AWS Direct Connect.

Q93**What is a Peering Connection in Azure?**

VNet Peering connects two VNets, whether in the same region or across regions, for direct communication, functioning similarly to AWS VPC Peering.

Q94**What is Azure Bastion?**

Azure Bastion provides secure RDP and SSH access to VMs directly through the Azure portal without needing to assign the VM a public IP at all, which meaningfully reduces the attack surface compared to opening RDP/SSH directly to the internet.

Q95**What is an Application Security Group (ASG)?**

ASGs let me group VMs logically for simplified NSG rule management -- instead of writing rules against individual IPs, I can write a rule against an application tier's ASG and add or remove VMs from that group without touching the NSG rules themselves.

Q96**What is an Azure Private Link?**

Private Link provides private connectivity to Azure services over a VNet without that traffic traversing the public internet, functionally the same purpose as AWS PrivateLink.

Q97**What are Subnets in Azure?**

Subnets segment a VNet to organize resources and apply distinct routing and access control rules to different tiers of an application, the same conceptual role subnets play in an AWS VPC.

Q98**What is an Azure Public IP Address?**

A Public IP resource allows an Azure resource to communicate directly with the internet, similar to an Elastic IP in AWS.

Q99**What is a Route Table in Azure?**

Route tables define custom routing rules that control traffic flow within a VNet, letting me override Azure's default system routes when traffic needs to go through a specific path, like a network virtual appliance.

Q100**What is Azure DNS?**

Azure DNS is Microsoft's managed domain hosting service, providing fast, highly available DNS resolution -- the Azure equivalent of Route 53's hosted zone functionality.

Q101 What is Azure Front Door?

Front Door is a global, application-layer load balancer and CDN combined, providing fast, secure routing to the closest healthy backend -- similar in role to AWS Global Accelerator combined with CloudFront.

Q102 What is a Service Endpoint in Azure?

A Service Endpoint extends a VNet's private address space to a specific Azure service, letting resources reach that service privately without the traffic leaving Azure's backbone network.

Q103 What is a DDoS Protection Plan in Azure?

It's Azure's managed protection against distributed denial-of-service attacks, providing enhanced mitigation capabilities beyond the basic protection every Azure resource gets by default.

Q104 What is Azure Monitor Network Insights?

It gives a unified, visual view of network health and topology across Azure resources, which speeds up troubleshooting by showing connectivity issues and traffic patterns without piecing them together from several separate tools.

Q105 What is a Network Virtual Appliance (NVA) in Azure?

An NVA is a VM that provides advanced networking functionality -- a firewall, a custom router -- beyond what native Azure networking constructs offer, and route tables are typically used to direct traffic through it for inspection or filtering.

06

VERSION CONTROL: GIT & GITHUB

Q106

What are the benefits of using version control systems like Git?

Collaboration without overwriting each other's work, a full history of every change with who made it and when, branching and merging so features can be developed independently, a remote backup so local loss isn't catastrophic, the ability to revert to any previous version, and pull-request-based code review before anything merges into the main codebase.

KEY POINTS TO MENTION

- Collaboration, tracked history, branching/merging, backup, version rollback, PR-based review

Q107

How do you resolve conflicts in Git?

Git flags the conflicting files, and I open each one to find the <<<<<<<, =====, >>>>>>> markers showing both versions. I edit the file to choose or combine the correct changes, remove the markers, stage the file with git add to mark it resolved, then complete the operation with git commit for a merge or git rebase --continue for a rebase, and finally push the resolved changes.

Q108

What is a rebase, and when would you use it instead of merging?

Rebase replays your commits on top of another branch's latest commits, producing a clean, linear history instead of the extra merge commits a regular merge creates. I use it to bring my feature branch up to date with main before opening a PR, but I avoid rebasing anything that's already been pushed and shared, since rewriting shared history causes real pain for collaborators.

Q109

Can you explain Git branching strategies (Git Flow, Trunk-Based Development)?

Git Flow uses several long-lived branches -- main for production, develop for ongoing work, feature branches for new work, release branches cut from develop, and hotfix branches from main for urgent fixes -- which suits teams with scheduled releases. Trunk-based development has developers committing small, frequent changes directly to a shared trunk with short-lived feature branches, which fits CI/CD-heavy teams shipping continuously. GitHub Flow and feature branching are lighter-weight variants in between, emphasizing short-lived branches and pull requests.

KEY POINTS TO MENTION

- Git Flow: main/develop/feature/release/hotfix — scheduled releases
- Trunk-based: frequent small commits to trunk — continuous deployment

Q110**How do you integrate GitHub with CI/CD tools?**

GitHub can fire webhooks to tools like Jenkins or GitLab CI on events like a commit or pull request, or use its native GitHub Actions for built-in automation. For third-party tools, I authenticate using a personal access token or a GitHub App, and I use Docker images pulled from Docker Hub within pipelines to keep build environments consistent, with CI results reported back as PR status checks before a merge is allowed.

07

GITHUB ACTIONS

Q111 What are GitHub Actions and how do they work?

GitHub Actions is a CI/CD tool that automates tasks directly within a repository, defined through YAML workflows in the `.github/workflows` directory. Workflows trigger on events like `push`, `pull_request`, or a schedule, and define jobs made up of steps that execute inside a virtual environment GitHub provisions.

Q112 How do you create a GitHub Actions workflow?

I add a YAML file under `.github/workflows/`, specifying on for the triggering event, jobs for the units of work, and steps within each job for individual actions -- checking out the repo, running a script, or invoking a pre-built action from the marketplace.

Q113 What are runners in GitHub Actions?

Runners are the servers that actually execute a workflow's jobs. GitHub-hosted runners come with common tooling pre-installed across Linux, macOS, and Windows, and I'd switch to self-hosted runners when a job needs to reach internal infrastructure or requires a specific environment GitHub-hosted runners don't provide.

Q114 How do you securely store secrets in GitHub Actions?

GitHub's Secrets feature encrypts values like API keys and credentials, accessible in a workflow via `${{ secrets.MY_SECRET }}` with the actual value automatically masked in logs so it never gets echoed out accidentally.

08

ARGOCD

Q115 What is ArgoCD and what problem does it solve?

ArgoCD is a declarative GitOps continuous delivery tool for Kubernetes that tracks and continuously synchronizes the live state of a cluster with the desired state defined in a Git repository, so Git becomes the actual single source of truth for what's running rather than just the starting point for a one-time deploy.

Q116 What are the key components of ArgoCD?

An Application resource defines what to deploy, which Git repo to pull from, and which cluster to target; the Repository is the Git source holding manifests, Helm charts, or Kustomize configs; Sync is the reconciliation process that brings the live cluster in line with Git; and ArgoCD's monitoring continuously shows the diff between desired and current state.

Q117 How do you perform application synchronization in ArgoCD?

Sync can be manual, triggered from the UI or via `argocd app sync APP_NAME`, or automatic, where ArgoCD continuously applies Git changes as they land, optionally with self-healing enabled so any manual drift in the cluster gets automatically reverted back to match Git.

Q118 What is the difference between GitOps and traditional CD pipelines?

Traditional CD pushes changes out to environments, meaning the pipeline itself needs write credentials to production. GitOps, as ArgoCD implements it, pulls changes instead -- an in-cluster agent continuously reconciles against Git -- which gives better visibility, tighter security since no external system needs direct write access, and automatic drift correction.

09

CI/CD PIPELINE

Q119 How would you design a CI/CD pipeline for a project?

Code commit triggers the pipeline: build with the right tool for the stack -- Maven, npm, pip -- automated testing including unit and integration tests, static code analysis with something like SonarQube or ESLint, packaging into an artifact like a JAR or Docker image, storing that artifact in a repository like Nexus or Docker Hub, deploying to staging for further testing, an approval gate before production, deployment to production, and post-deployment monitoring with Grafana and Prometheus to confirm the release is stable.

KEY POINTS TO MENTION

- Commit → build → test → static analysis → package → store artifact → staging → approval → prod → monitor

Q120 What tools have you used for CI/CD, and why did you choose them?

Jenkins for its flexibility and huge plugin ecosystem across almost any tech stack; GitHub Actions for smaller projects or where deep GitHub integration matters; GitLab CI when the codebase is already hosted on GitLab, for the seamless built-in integration; ArgoCD specifically for GitOps-based delivery into Kubernetes; Docker for consistent packaging across environments; and Terraform for automating the infrastructure the pipeline deploys into.

Q121 Can you explain the different stages of a CI/CD pipeline?

Source stage where code is committed to version control; build stage where source compiles into an executable or package; test stage running unit, integration, and performance tests; artifact stage where the build becomes a deployable unit stored in a repository; deployment stage pushing to staging and then production after approval; and post-deployment monitoring to confirm stability.

Q122 What are artifacts, and how do you manage them in a pipeline?

Artifacts are the actual build outputs -- JAR/WAR files, Docker images, zip packages, binaries. I manage them by storing them in a repository like Nexus, Artifactory, or Docker Hub, versioning and tagging each one based on the release or build number for traceability and rollback, and applying retention policies so old, unused artifacts don't accumulate indefinitely.

Q123**How do you handle rollbacks in the case of a failed deployment?**

With blue-green or canary deployments, rollback is just redirecting traffic back to the previous version without downtime. Since artifacts are versioned, I can also redeploy the last known-good version directly from the artifact repository. Automated health checks post-deployment can trigger this rollback automatically, and for infrastructure-level failures, Terraform lets me revert to a previous infrastructure state as well.

KEY POINTS TO MENTION

- Canary/Blue-Green: redirect traffic back
- Versioned artifacts enable redeploying last known-good build
- Automated health-check-triggered rollback
- Terraform for infrastructure-level reverts

10

CONTAINERIZATION: DOCKER & KUBERNETES

Q124

What is Docker, and how does it differ from a virtual machine?

Docker packages an application with its dependencies into a container that shares the host OS kernel but isolates processes, filesystems, and resources -- lightweight and fast to start. A VM emulates full hardware and runs a complete separate OS on a hypervisor, giving stronger isolation but at much higher resource cost and slower startup. I default to containers for microservices and reserve full VMs for cases needing stronger isolation or a fundamentally different OS than the host.

KEY POINTS TO MENTION

- Performance: containers lightweight/fast, VMs heavier
- Isolation: VMs stronger (hardware-level), containers process-level
- Resource efficiency: containers use less CPU/memory

Q125

How do you create and manage Docker images and containers?

I write a Dockerfile specifying the base image, copying application code, installing dependencies, and setting the entry point, then build it with `docker build -t my-app:1.0 .` and push it to a registry with `docker push`. To manage running containers I use `docker run` to start one, `docker stop/start` to control its lifecycle, `docker rm` to remove it, and Docker Compose when I need to define and run multiple containers together.

Q126

How do you optimize Docker images for production?

Smaller base images like alpine to cut size and reduce attack surface, multi-stage builds so build tools and dependencies never make it into the final image, minimizing layers by combining commands where it makes sense, a `.dockerignore` file to keep unnecessary files out of the build context, and ordering Dockerfile instructions to take advantage of layer caching -- rarely-changing steps like dependency installation before frequently-changing ones like copying source code.

KEY POINTS TO MENTION

- Small base images, multi-stage builds, minimal layers, `.dockerignore`, cache-aware instruction ordering

Q127

What is Kubernetes, and how does it help in container orchestration?

Kubernetes automates the deployment, scaling, and management of containerized applications -- it handles automatic scaling based on traffic or resource usage, load balancing across containers, self-healing by restarting failed containers and killing unresponsive ones, automated rollouts and rollbacks with zero downtime, and resource allocation across the cluster.

Q128**Explain how you've set up a Kubernetes cluster.**

For a managed setup I'd use a cloud provider's offering like EKS, GKE, or AKS, letting the provider handle the control plane. For a self-managed cluster, I'd use kubeadm init to set up the control plane node and kubeadm join to add worker nodes, install a CNI networking plugin like Calico or Weave immediately since pods can't communicate without one, then deploy applications as Deployments, Services, and ConfigMaps, and finally set up Prometheus and Grafana for cluster monitoring.

Q129**What are Kubernetes services, and how do they differ from Pods?**

Pods are the smallest unit in Kubernetes, representing one or more tightly coupled containers sharing network and storage, and they're ephemeral -- they come and go as deployments roll out or nodes fail. A Service provides a stable IP address and DNS name for a set of Pods identified by a label selector, so traffic keeps reaching healthy Pods even as the underlying Pods themselves are replaced.

1 1

KUBERNETES

Q130 What is Kubernetes and why is it used?

Kubernetes is an open-source container orchestration platform that automates deploying, scaling, and managing containerized applications across a cluster of servers, letting distributed applications run efficiently without manual, error-prone intervention for every scaling or failover event.

Q131 What are Pods in Kubernetes?

A Pod is the smallest deployable object in Kubernetes, representing a single instance of a running process that can contain one or more tightly coupled containers sharing the same network namespace and storage.

Q132 Explain the difference between a Deployment and a StatefulSet in Kubernetes.

A Deployment is for stateless applications and ensures the correct number of interchangeable Pods are running, scaling and recreating them freely. A StatefulSet is for stateful applications needing stable network identity and persistent storage per Pod, with ordered, predictable deployment and scaling -- useful for databases and other services where each replica has its own identity and data.

Q133 How do you expose a Kubernetes application to external traffic?

A Service of type LoadBalancer provisions a cloud load balancer for the app; Ingress provides HTTP/HTTPS routing with support for features like SSL termination and host/path-based rules; and NodePort exposes the app on a static port on every node, which I mostly use for quick testing rather than production.

Q134 How does Kubernetes handle storage?

PersistentVolumes represent actual durable storage resources in the cluster, PersistentVolumeClaims are requests from a Pod for storage matching certain criteria, and a StorageClass defines the type of storage -- SSD versus HDD, for instance -- enabling dynamic provisioning so a PV is created automatically to satisfy a PVC instead of needing to be pre-provisioned manually.

1 2

KUBERNETES ARCHITECTURE

Q135

What are the main components of Kubernetes architecture?

The Control Plane manages the overall cluster -- API Server, etcd, Scheduler, and Controller Manager -- and Worker Nodes run the actual containerized applications through the Kubelet, Kube-proxy, and container runtime.

KEY POINTS TO MENTION

- Control Plane: API Server, etcd, Scheduler, Controller Manager
- Worker Nodes: Kubelet, Kube-proxy, container runtime

Q136

What is the role of the Kubernetes API Server?

The API Server is the front-end to the entire control plane, exposing the Kubernetes API. It processes every REST request -- kubectl commands or otherwise -- updates the cluster's state accordingly, and manages communication between internal control plane components and external users.

Q137

What is etcd and why is it important in Kubernetes?

etcd is the distributed key-value store holding all cluster state -- Pods, Secrets, ConfigMaps, Services, everything. It's the actual source of truth for cluster configuration, and it ensures data consistency and high availability across control plane nodes, which is why losing etcd quorum is one of the most serious things that can happen to a cluster.

Q138

What does the Kubernetes Scheduler do?

The Scheduler assigns Pods to nodes, factoring in resource availability, node conditions, affinity/anti-affinity rules, and other constraints, aiming to distribute Pods efficiently across the cluster rather than overloading any single node.

Q139

What is a Kubelet, and what role does it play?

The Kubelet is the agent running on every worker node, making sure the containers described in Pod specs are actually running correctly, communicating with the control plane to receive instructions and report node/pod status, and interacting with the container runtime to manage container lifecycle.

Q140

What is a pod in Kubernetes?

A Pod is the smallest and simplest Kubernetes object -- a group of one or more containers sharing storage and network resources and the same execution context, typically running a single instance of an application.

Q141 How does Kubernetes networking work?

Kubernetes uses a flat network model where every Pod gets its own unique IP, and Pods can communicate across nodes without NAT. CNI plugins like Calico, Flannel, or Weave implement that connectivity, and kube-proxy on each node manages Service networking, routing traffic to the correct backing Pods.

Q142 What is the role of the Controller Manager?

The Controller Manager runs a set of controllers that continuously monitor cluster state and reconcile it toward the desired state -- the Node Controller watches node health, the Replication Controller ensures the right number of Pod replicas are running, the Job Controller manages job completion, and so on.

Q143 What is the role of the Kube-proxy?

Kube-proxy maintains network rules on every worker node, routing traffic from Services to the correct backing Pods across nodes, typically implemented through iptables or IPVS rules for efficient request routing.

Q144 What are Namespaces in Kubernetes?

Namespaces divide cluster resources between multiple users or teams, organizing objects, separating resources for different environments or teams, and letting resource limits and access controls be applied at the namespace level rather than only cluster-wide.

Q145 How does Kubernetes achieve high availability?

Multiple control plane nodes so losing one doesn't take down cluster management, a clustered, highly available etcd for consistent state and failover, and Pod replication across multiple worker nodes so a single node failure doesn't take a service down entirely.

Q146 What is the function of the Cloud Controller Manager?

It manages cloud-provider-specific logic -- provisioning load balancers, managing node instances, handling persistent storage integration -- keeping that vendor-specific logic decoupled from the core Kubernetes components.

Q147 What is the significance of a Service in Kubernetes?

A Service defines a logical set of Pods and a policy to access them, providing a stable IP address and DNS name even as the underlying Pods are dynamically created or destroyed -- exposing an app either internally via ClusterIP or externally via a LoadBalancer-type Service.

Q148 How does Kubernetes handle scaling?

Manual scaling with `kubectl scale` to directly adjust replica count; Horizontal Pod Autoscaler, which automatically scales replica count based on CPU/memory or custom metrics; and Vertical Pod Autoscaler, which adjusts a Pod's resource requests and limits based on observed consumption rather than the number of replicas.

1 3

INGRESS CONTROLLER

Q149 What is an Ingress Controller in Kubernetes?

An Ingress Controller is a specialized load balancer for the cluster that interprets Ingress resources and routes external HTTP/S traffic to the right services based on host and path rules -- NGINX, Traefik, and HAProxy are common ones I've worked with.

Q150 How does an Ingress Controller differ from a Load Balancer?

An Ingress Controller is purpose-built for HTTP/S traffic and sophisticated routing based on host or path, while a generic Load Balancer distributes traffic across instances and can handle traffic types beyond HTTP. In practice they work together -- a cloud load balancer often sits in front of the Ingress Controller, which then does the finer-grained routing.

Q151 Can you explain how to set up an Ingress Controller in a Kubernetes cluster?

Choose a controller like NGINX, deploy it via a Helm chart or manifest into the cluster, then define Ingress resources specifying host and path rules pointing at backend Services, and finally update DNS to point at the Ingress Controller's external IP or load balancer endpoint.

Q152 What are some common features of an Ingress Controller?

Path-based and host-based routing, TLS termination with SSL certificate management, load balancing across backend Services, integration with external authentication and authorization, and rate limiting or caching for traffic control.

KEY POINTS TO MENTION

- Path/host-based routing, TLS termination, load balancing, auth integration, rate limiting/caching

Q153 How do you handle SSL termination with an Ingress Controller?

I store the TLS certificate and key in a Kubernetes Secret and reference it in the Ingress resource's tls section, specifying the host it applies to -- the Ingress Controller then handles decrypting HTTPS traffic at that layer before forwarding it to the backend Service.

Q154 What are some best practices when configuring an Ingress Controller?

Always use TLS for anything public-facing, keep Ingress rules simple rather than over-engineering routing logic, implement monitoring and logging for troubleshooting, use annotations for controller-specific configuration like timeouts, and add rate limiting to protect backend services from being overwhelmed.

Q155**How do you troubleshoot issues with an Ingress Controller?**

Check the Ingress resource configuration is correct and pointing at the right Service and port, inspect the Ingress Controller pod's logs for errors, test connectivity directly with curl, verify DNS actually points at the controller's external IP, and confirm the backend Service has healthy endpoints at all.

Q156**What is the role of annotations in an Ingress resource?**

Annotations configure controller-specific behavior beyond the standard Ingress spec -- load balancing algorithm, SSL settings, rate limits, custom rewrite rules -- and exactly which annotations are available depends on which Ingress Controller is actually implementing the resource.

Q157**Can you explain what a Virtual Service is in the context of Ingress Controllers?**

A Virtual Service, from service mesh technologies like Istio, defines finer-grained request routing, traffic splitting, and service-level policies within the mesh -- more advanced than what a standard Ingress resource provides, which handles the external entry point rather than internal service-to-service routing.

Q158**How do you secure your Ingress Controller?**

TLS on all traffic, authentication mechanisms like OAuth or JWT where appropriate, network policies restricting access to the controller itself, rate limiting to guard against DDoS-style abuse, and keeping the controller updated to the latest stable version to avoid known vulnerabilities.

1 4

INFRASTRUCTURE AS CODE: TERRAFORM & ANSIBLE

Q159

What is Infrastructure as Code (IaC), and how does it benefit a DevOps environment?

IaC manages and provisions infrastructure through machine-readable configuration files instead of manual, interactive setup. It gives consistency across environments, efficiency through automation instead of manual intervention, easy scalability by replicating components, version control and audit trails for infrastructure just like application code, and a shared language for collaboration between teams.

KEY POINTS TO MENTION

- Consistency, efficiency, scalability, version control, collaboration

Q160

How do you manage cloud infrastructure with Terraform?

I define infrastructure in .tf configuration files describing the desired resource state, run terraform init to set up the working directory and download providers, terraform plan to preview exactly what will change, and terraform apply to provision it. Updates are just changing the config and running apply again, and terraform destroy tears resources down cleanly when they're no longer needed.

Q161

Can you explain the difference between Terraform and Ansible?

Terraform focuses on provisioning and managing infrastructure lifecycle end to end -- create, modify, delete -- using declarative HCL configuration across cloud providers. Ansible is primarily a configuration management tool, automating software deployment and configuration on servers that already exist, using procedural YAML playbooks, and it doesn't manage infrastructure provisioning the way Terraform does.

Q162

How do you handle versioning in Infrastructure as Code?

I store all IaC files in Git, use meaningful commit messages and tags to mark meaningful changes and versions, use release branches or tags to manage configuration differences across environments, and integrate that versioned IaC with CI/CD pipelines so testing, deployment, and rollback are automated based on the versioned configuration rather than manual application.

Q163

What challenges have you faced with configuration management tools?

Complexity managing large-scale infrastructure and dependencies as playbooks or recipes grow, consistency across different environments with varying OS versions or package dependencies, scalability as infrastructure grows or changes, securely handling sensitive credentials within the tool, and integrating with existing systems already in the organization's ecosystem. Addressing these comes down to modular design, thorough testing, and careful planning rather than any single fix.

1 5

CLOUD COMPUTING: AWS

Q164

What cloud platforms and AWS services have you worked with?

I've worked extensively with EC2 for scalable compute, S3 for object storage, RDS for managed relational databases, Lambda for serverless workloads, VPC for network isolation, CloudFormation and Terraform for infrastructure as code, and EKS for managing Kubernetes clusters -- the core services that make up most real-world AWS architectures I've built or maintained.

Q165

How do you ensure high availability and scalability in the cloud?

For high availability: deploying across multiple availability zones for redundancy, Elastic Load Balancing to distribute traffic, and Auto Scaling Groups to automatically adjust instance count based on demand. For scalability: horizontal scaling by adding or removing instances, leveraging managed services like RDS read replicas or DynamoDB for database scalability, and caching with ElastiCache to reduce database load and improve response times.

KEY POINTS TO MENTION

- HA: Multi-AZ, ELB, Auto Scaling Groups
- Scalability: horizontal scaling, RDS read replicas/DynamoDB, ElastiCache

Q166

What are the best practices for securing cloud infrastructure?

IAM roles and policies following least privilege, encryption for data at rest and in transit, security groups and NACLs to control traffic along with AWS WAF for web application protection, CloudTrail and CloudWatch for logging and monitoring account activity, and regular security audits to catch vulnerabilities and confirm ongoing compliance.

Q167

Can you explain how to set up auto-scaling for an application?

Launch and configure an EC2 instance as the template, create a Launch Template or Configuration capturing the AMI, instance type, security groups, and user data, create an Auto Scaling Group referencing that template across chosen subnets and AZs, define scaling policies -- target tracking, step scaling, or scheduled -- based on metrics like CPU utilization, attach a load balancer if traffic needs to be distributed across instances, and monitor with CloudWatch to fine-tune the policies over time.

KEY POINTS TO MENTION

- Launch template → ASG → scaling policy (target tracking/step/scheduled) → attach LB → monitor & tune

What is the difference between IaaS, PaaS, and SaaS?

IaaS, like EC2, gives virtualized compute and networking while I manage the OS, runtime, and application myself. PaaS, like Elastic Beanstalk, manages the OS and runtime for me so I just deploy application code. SaaS, like Salesforce or Office 365, is fully managed software I simply use, with no infrastructure or platform management on my end at all.

KEY POINTS TO MENTION

- IaaS: EC2 — full control, most responsibility
- PaaS: Elastic Beanstalk — deploy code only
- SaaS: fully managed application, no infra/platform responsibility

Q169

What is Azure DevOps?

Azure DevOps is Microsoft's cloud-based suite covering the full software development lifecycle -- Azure Repos for version control, Azure Pipelines for CI/CD, Azure Boards for project management, Azure Artifacts for package management, and Azure Test Plans for testing, all integrated into one platform.

Q170

What are the key components of Azure DevOps?

Azure Boards for Kanban-style project tracking, Azure Repos for Git-based source control, Azure Pipelines for build and release CI/CD, Azure Test Plans for managing tests and defects, and Azure Artifacts for managing and sharing packages across projects.

Q171

What is CI/CD, and how does Azure DevOps support it?

CI automatically integrates and tests code changes, while CD automates releasing that code to production. Azure Pipelines implements both, letting me build workflows that compile, test, and deploy applications across environments with minimal manual intervention.

Q172

How do you implement CI/CD using Azure Pipelines?

Define a pipeline using YAML or the visual designer, connect it to the source repository, define build steps for compiling and testing, set up release pipelines to deploy to various environments, and configure triggers so builds kick off automatically on commits or pull requests.

Q173

What is Azure Resource Manager (ARM)?

ARM is Azure's deployment and management service providing a consistent layer for creating, updating, and deleting resources, using ARM templates to define infrastructure and configuration declaratively.

Q174

What is Infrastructure as Code (IaC) in Azure DevOps?

IaC in Azure DevOps means defining and provisioning Azure resources programmatically using ARM templates, Terraform, or Azure Bicep instead of manual portal clicks, giving versioning, repeatability, and easier long-term infrastructure management.

Q175 How do you manage secrets in Azure DevOps?

Azure Key Vault securely stores secrets like passwords and API keys, and the Azure DevOps library can hold variable groups containing secrets referenced in pipelines -- either way, permissions need to be configured carefully so only authorized users and services can actually access them.

Q176 What are the different types of pipelines in Azure DevOps?

Build Pipelines automate compiling code, running tests, and producing artifacts, while Release Pipelines manage deploying those artifacts to environments like staging or production.

Q177 Can you explain the role of Azure Boards in Agile development?

Azure Boards provides Kanban boards, backlogs, sprint planning, and reporting for Agile teams, letting them manage user stories, tasks, and bugs collaboratively with full visibility throughout the development process.

Q178 What are Azure DevOps Service Hooks?

Service Hooks integrate Azure DevOps with other services by triggering events on changes -- sending a Slack notification or triggering a build in Jenkins or GitHub, for example, whenever something like a pull request or completed build happens in Azure DevOps.

Q179 What are Azure DevOps Extensions, and how do you use them?

Extensions add functionality to Azure DevOps -- integrating third-party tools or adding new features -- available through the Azure DevOps Marketplace. I install and configure them as needed, and they immediately become usable features within the existing Azure DevOps environment.

Q180 How do you implement security in Azure DevOps?

User and group permission management to control access, service connections to securely link to external services without exposing raw credentials, Azure Active Directory integration for identity management, and branch policies with pull request approvals to enforce code quality and security checks before merges.

Q181 What is a Release Gate in Azure DevOps?

A Release Gate is a condition that must be satisfied before a release proceeds to the next stage -- querying an external system, waiting for manual approval, checking a prior deployment's status -- ensuring releases meet specific criteria before continuing.

Q182 How can you implement monitoring in Azure DevOps?

Azure Monitor and Application Insights provide the monitoring layer -- setting up alerts for failures or performance degradation, giving a unified view of applications, infrastructure, and networks, with Application Insights specifically focused on application performance and user behavior.

Q183 What are YAML Pipelines, and how do they differ from Classic Pipelines?

YAML Pipelines are defined in a file checked into the source repo, giving version control and easier collaboration, while Classic Pipelines use a visual designer in the portal. YAML Pipelines are more flexible, reusable, and versioned alongside the application, which is why I default to them for anything beyond a quick proof of concept.

Q184 Explain the concept of Branch Policies in Azure Repos.

Branch Policies enforce rules on how changes reach a branch -- requiring pull requests, code review, or a successful build before merging -- which maintains code quality and ensures changes are properly reviewed before landing.

Q185 What is Azure Container Registry (ACR), and how does it integrate with Azure DevOps?

ACR is a managed private Docker registry, and it integrates with Azure Pipelines so container images are built, pushed, and pulled as part of CI/CD -- a pipeline typically publishes to ACR right after a successful build.

Q186 How do you handle environment variables in Azure DevOps Pipelines?

Pipeline variables at the pipeline or stage level, Variable Groups for values reused across multiple pipelines, the `env` keyword for setting variables directly within a YAML job, and Azure Key Vault references for anything sensitive that needs to be securely pulled in.

Q187 What is the purpose of using Approval Gates in Azure DevOps?

Approval Gates enforce manual sign-off before a stage proceeds, adding oversight so deployments are verified against specific criteria by relevant stakeholders, minimizing the risk of an unreviewed change reaching production.

Q188 What is the significance of using Personal Access Tokens (PAT) in Azure DevOps?

PATs authenticate to Azure DevOps without a username and password, which is especially useful for scripting and automation. They can be scoped to specific permissions, so managing and securing them carefully -- narrow scope, reasonable expiration -- matters to avoid unauthorized access.

Q189 What are the benefits of using Azure DevTest Labs?

Quick environment setup for testing and development, cost management features to track and cap spending, reusable templates for consistent environments, and integration with Azure DevOps for automated deployments into those lab environments.

Q190 How do you create and manage Azure DevOps service connections?

In Project Settings under Pipelines, I create a new service connection, choose the type -- Azure Resource Manager, GitHub, and so on -- provide the authentication details and permissions needed, and then reference that connection in pipelines for secure access to the external resource.

Q191**What is a build artifact in Azure DevOps?**

A build artifact is a file or set of files produced by a build -- compiled binaries, packages, resources needed for deployment -- that Azure DevOps lets you publish and share across pipeline stages or between pipelines.

Q192**How do you automate tests in Azure DevOps?**

Azure Test Plans for manual and automated test management, unit testing frameworks like NUnit or JUnit integrated directly into the build pipeline, continuous testing features that run automated tests during build or release, and publishing results and reports within Azure DevOps for tracking.

Q193**What is the purpose of Azure Monitor, and how does it work with Azure DevOps?**

Azure Monitor provides visibility into application performance, infrastructure health, and usage metrics, and combined with Azure DevOps and Application Insights, gives teams the insight and alerting needed to catch issues and improve reliability throughout the development lifecycle.

1 7

MONITORING & LOGGING

Q194

How do you monitor the health of a system in production?

I track key metrics like CPU, memory, disk, response times, error rates, and throughput, run uptime checks against real application endpoints rather than shallow pings, continuously review logs for warnings and errors, set threshold-based alerts to get notified in real time, and keep dashboards that show overall system health at a glance.

KEY POINTS TO MENTION

- Key metrics, uptime checks, log review, threshold alerts, dashboards

Q195

What tools have you used for monitoring?

Prometheus for time-series metric collection and PromQL querying, Grafana for visualizing those metrics through dashboards, Alertmanager paired with Prometheus for routing alerts to Slack or email, the ELK stack for log aggregation and analysis, and the Prometheus Operator specifically for Kubernetes cluster monitoring.

Q196

How do you set up alerts for monitoring systems?

Prometheus rules define thresholds -- CPU usage above 80%, for example -- routed through Alertmanager to the right channel. I set threshold-based alerts for response time and error rate, custom application-specific alerts for things like failed transactions, Kubernetes readiness/liveness probes to catch unhealthy services, and Grafana's own alerting for anything visualized there.

Q197

Can you explain the ELK stack and how you've used it?

Elasticsearch stores and searches large volumes of log data, Logstash collects and processes logs from different sources before shipping them to Elasticsearch, and Kibana provides the visualization and search interface on top. I've used it to aggregate logs across microservices, filtering and formatting them in Logstash, then building Kibana dashboards to monitor error rates, latency, and overall service health.

Q198

How do you troubleshoot an application using logs?

Centralize all logs in one place first, then search specifically for errors or exceptions in the relevant time window, trace a request through multiple services by correlating request or user IDs, examine the context leading up to the error -- resource constraints, failed dependencies -- filter by severity to focus on what matters, and rely on consistent structured log formats so parsing and searching stays easy.

Q199

How do you integrate security into the DevOps lifecycle (DevSecOps)?

During planning, security requirements and threat modeling are considered up front. During coding, secure coding practices and code analysis catch issues early. During build, automated static analysis runs in CI. During test, vulnerability scanning covers the application and infrastructure. At deploy, IaC scanners check for misconfigurations. And in operation, continuous monitoring and alerting catch anomalies, with automated incident detection closing the loop.

KEY POINTS TO MENTION

- Plan → code → build → test → deploy → operate → monitor, security embedded at every stage

Q200

What tools have you used to scan for vulnerabilities?

OWASP Dependency-Check for known vulnerabilities in third-party libraries against the NVD database, SonarQube for static code analysis catching code smells, vulnerabilities, and bugs, Trivy for scanning container images, filesystems, and Git repositories, Aqua Security or Clair for container image scanning, Snyk for open-source library and Docker image vulnerabilities integrated directly into CI/CD, Checkmarx for SAST on source code, and checkov or terrascan specifically for scanning Terraform IaC for misconfigurations. Wiring all of these into the pipeline is what actually gives a genuine shift-left security posture, catching issues from code through deployment.

1 9

JENKINS

Q201 What is Jenkins? Why is it used?

Jenkins is an open-source automation server used primarily for continuous integration and continuous delivery -- automating build, test, and deployment so bugs are caught early, software quality improves, and delivery time shrinks.

Q202 How does Jenkins achieve Continuous Integration?

Jenkins integrates with version control systems like Git, automatically triggering builds and tests whenever changes are committed -- running unit tests, static analysis, and deploying if everything passes, with notifications sent to the team about build status along the way.

Q203 What is a Jenkins pipeline?

A Jenkins pipeline is a suite of plugins supporting implementing and integrating continuous delivery pipelines directly into Jenkins, letting me define complex build, test, and deployment workflows as code rather than manually configured job steps.

Q204 What are the two types of Jenkins pipelines?

Declarative Pipeline, a newer, simpler syntax defined within a pipeline block, and Scripted Pipeline, written in full Groovy syntax for more flexibility at the cost of added complexity.

Q205 What is the difference between a freestyle project and a pipeline project in Jenkins?

A Freestyle Project is the basic Jenkins job type for simple tasks like running a shell script or a build step. A Pipeline Project defines complex job sequences, orchestrating multiple builds, tests, and deployments across environments as code.

Q206 How do you configure a Jenkins job to be triggered periodically?

Enable the "Build periodically" option in the job configuration and define the schedule using cron syntax -- something like `H/5 * * * *` to run every 5 minutes.

Q207 What are the different ways to trigger a build in Jenkins?

Manual trigger via "Build Now", triggering through source code changes via Git hooks, a cron schedule for periodic builds, webhooks or API calls, and triggering a build after another build completes.

Q208 What are Jenkins agents? How do they work?

Agents, also called nodes or slaves, are machines configured to execute jobs on behalf of the Jenkins controller/master. The controller delegates work to agents, which can run on different platforms, distributing build load across multiple machines instead of everything running on the controller.

Q209 How can you integrate Jenkins with other tools like Git, Maven, or Docker?

Through plugins -- the Git plugin to pull code from a repository, the Maven plugin for building Java projects, and the Docker plugin for building and deploying containers, all configured within a job or pipeline.

Q210 What is Blue Ocean in Jenkins?

Blue Ocean is a modern, more user-friendly Jenkins interface offering a simplified, visual view of CI/CD pipelines, making it easier to understand pipeline flow and troubleshoot failures than the classic Jenkins UI.

Q211 What are the steps to secure Jenkins?

Enable Matrix-based or Role-based access control, run Jenkins behind a secure network with HTTPS, use SSH keys for secure communication, install security-relevant plugins like OWASP Dependency-Check, and keep Jenkins and all its plugins up to date to avoid known vulnerabilities.

Q212 What is a Jenkinsfile?

A Jenkinsfile is a text file defining a Jenkins pipeline, versioned alongside the code it builds, used to automate build, test, and deployment -- written as either a declarative or scripted pipeline.

Q213 How does Jenkins handle parallel execution in pipelines?

The parallel directive lets multiple stages run simultaneously -- for example running unit tests and integration tests concurrently instead of sequentially -- which reduces overall build time when those stages don't depend on each other.

Q214 How can you monitor Jenkins logs and troubleshoot issues?

Jenkins logs are visible through the UI's "Manage Jenkins" → "System Log" section, individual job-specific logs are in each job's build history, and for deeper detail I check the actual server log files on the machine hosting Jenkins.

Q215 How can you handle failed builds in Jenkins?

Configure automatic retries a specified number of times after a failure, set up post-build actions like notifications or triggering other jobs on failure, and use conditional logic in pipelines -- like try-catch blocks -- to handle failures gracefully instead of letting the whole pipeline crash unhelpfully.

Q216 What is a symlink in Linux?

A symlink is a file that points to another file or directory, acting as an indirect reference -- if the target moves or is deleted, the symlink breaks, which is worth remembering when scripts depend on one.

Q217 Explain the difference between a process and a daemon in Linux.

A process is a running instance of a program identified by a unique PID. A daemon is a background process, typically started at boot, that runs continuously performing a specific ongoing task -- things like sshd or cron.

Q218 How do you check the free disk space in Linux?

df shows disk space usage across mounted filesystems, and df -h gives that same output in a human-readable format -- that's the first thing I check when a server is behaving oddly and disk pressure is a suspect.

Q219 What is SSH and how is it useful in a DevOps context?

SSH is a cryptographic protocol for secure communication between machines, and it's fundamental to DevOps work -- remote server access, executing commands remotely, and securely transferring files all run through it.

Q220 Explain the purpose of the grep command in Linux.

grep searches for specific patterns within files or command output, letting me extract exactly the relevant lines by matching regular expressions or plain strings -- it's one of the tools I reach for constantly when digging through logs.

Q221 How would you find all files modified in the last 7 days in a directory?

find /path/to/directory -mtime -7 uses the -mtime option to filter by modification time within the last 7 days.

Q222 What is a shell script? Give an example of how you might use it in DevOps.

A shell script automates a sequence of commands for a shell interpreter like Bash. I use them for things like deploying an application, applying server configuration changes, or scheduling routine backups -- anywhere a repeatable sequence of commands would otherwise be run manually.

Q223 How can you manage software packages in Ubuntu/Debian-based systems?

apt commands like apt-get or apt-cache handle installing, removing, updating, or searching for packages -- sudo apt-get install being the most common one.

Q224 Explain the purpose of the chmod command in Linux.

chmod changes file or directory permissions, controlling read, write, and execute access separately for the owner, group, and others.

Q225 What is the role of cron in Linux?

cron is a time-based job scheduler that runs tasks automatically at specified times or intervals -- I use it for scheduling maintenance tasks, backups, and automated scripts that need to run on a recurring schedule.

Q226 What are runlevels in Linux, and how do they affect system startup?

Runlevels define which services run in a given operational mode -- single-user, multi-user, reboot/shutdown. Modern systemd-based systems have largely replaced runlevels with targets like multi-user.target and graphical.target instead.

Q227 How do you secure a Linux server?

Keep the system patched with regular updates, use a firewall like ufw or iptables to restrict access, enforce SSH security by disabling root login and requiring key-based authentication, install fail2ban to block repeated failed login attempts, and monitor logs while restricting permissions on sensitive files with chmod and chown.

KEY POINTS TO MENTION

- Patch regularly, firewall, SSH hardening, fail2ban, log monitoring, tight file permissions

Q228 What is LVM, and why is it useful in DevOps?

LVM lets me manage disks flexibly by creating logical volumes that can span multiple physical disks, enabling dynamic resizing and snapshots -- genuinely useful in environments that need to scale storage needs on the fly, like cloud infrastructure.

Q229 How do you monitor system performance in Linux?

top or htop for real-time CPU, memory, and process usage; vmstat for system performance stats; iostat for disk I/O; netstat or ss for network connections and traffic; and sar from the sysstat package for comprehensive, historical performance data.

Q230 What is the difference between a hard link and a soft link (symlink)?

A hard link shares the same inode as the original file, so the underlying data persists as long as any hard link to it still exists. A soft link just points to a file path, so it breaks if the target is moved or deleted.

Q231**How would you troubleshoot a Linux system that is running out of memory?**

Check memory usage with `free -h` or `vmstat`, identify the memory-hogging process with `top` or `htop`, review swap usage with `swapon -s`, check for memory leaks with `ps aux --sort=-%mem` or `smem`, and analyze `dmesg` for kernel-level memory issues or OOM killer activity.

Q232**Explain how you can schedule a one-time task in Linux.**

The `at` command schedules a one-time task -- `echo "sh backup.sh" | at 02:00` runs `backup.sh` at 2 AM. `atq` shows pending jobs, and `atrm` removes them.

Q233**How would you optimize a Linux system for performance?**

Disable unnecessary services with `systemctl`, tune kernel parameters via `sysctl`, monitor and manage disk I/O with `iostat` and use faster storage where it matters, adjust swappiness to control how aggressively the system swaps, and use profiling tools like `perf` to actually find bottlenecks instead of guessing.

Q234**How would you deal with high CPU usage on a Linux server?**

Identify the culprit process with `top` or `htop`, adjust its priority with `nice` or `renice` if appropriate, check whether the load is genuinely CPU-bound versus stemming from high I/O or memory pressure, review system logs for related errors, and optimize or tune the specific application if it turns out to be the actual root cause.

Q235**Explain how Linux file permissions work (rwx).**

Permissions split into owner, group, and others, each with read, write, and execute bits -- something like `rw-r--r--` means the owner has full access, the group can read and execute, and everyone else can only read.

Q236**What is the systemctl command, and why is it important for a DevOps engineer?**

`systemctl` controls `systemd`, the service manager on modern Linux distributions -- starting, stopping, restarting services, checking status, and managing boot targets. It's essential for managing critical infrastructure components like web servers and databases running as `systemd` services.

Q237**What is the purpose of iptables in Linux?**

`iptables` is a command-line firewall utility for configuring packet filtering, NAT, and routing rules -- used in DevOps to secure systems by controlling exactly what traffic is allowed in and out based on defined rules.

Q238**How would you handle logging in Linux?**

System logs live in `/var/log/`, managed with `rsyslog` or `syslog` for centralized logging, `journalctl` for viewing and filtering logs on `systemd`-based systems, `logrotate` for rotating and compressing large log files, and integration with a stack like ELK or Grafana Loki for real-time log visualization and analysis at scale.

Q239**What is a kernel panic, and how would you troubleshoot it?**

A kernel panic is a system crash from an unrecoverable kernel error. I'd check `/var/log/kern.log` or `journalctl` for the messages leading up to the panic, use `dmesg` to look for hardware or driver issues, and consider memory testing or reviewing recent kernel updates as likely culprits.

Q240**How do you install a specific version of a package in Linux?**

On Debian/Ubuntu, `apt-cache policy` lists available versions and `sudo apt-get install =` installs a specific one. On Red Hat/CentOS, `yum --showduplicates list` shows available versions and `sudo yum install -` installs it.

2 1

SONARQUBE

Q241 What is SonarQube, and why is it used?

SonarQube is an open-source platform that continuously inspects code quality, detecting bugs, vulnerabilities, and code smells across many languages, integrating tightly with CI/CD pipelines so teams improve code quality through consistent static analysis rather than relying on manual review alone.

Q242 What are the key features of SonarQube?

Code quality management tracking bugs, vulnerabilities, and code smells; security hotspot detection for risks like SQL injection or XSS; technical debt management estimating the time to fix detected issues; CI/CD integration with Jenkins, GitHub Actions, GitLab CI; custom quality profiles for project-specific rules; and support for over 25 programming languages.

Q243 How does SonarQube work in a CI/CD pipeline?

The SonarQube Scanner runs during the build phase, analyzing source code and sending results back to the SonarQube server, which generates a report of issues. The pipeline can be configured to fail if the defined quality gate isn't met, blocking poor-quality code from being released.

Q244 What are SonarQube Quality Gates?

A Quality Gate is a set of conditions -- around bugs, vulnerabilities, coverage, duplication -- that a project must meet to be considered acceptable. Configuring the pipeline to fail when the gate isn't met is what actually enforces the standard rather than just reporting on it.

Q245 What is a code smell in SonarQube?

A code smell is a maintainability issue that doesn't necessarily cause bugs or vulnerabilities but makes code harder to read, maintain, or extend -- long methods, too many parameters, poor variable naming are typical examples.

Q246 What is the difference between bugs, vulnerabilities, and code smells in SonarQube?

Bugs are issues likely to cause incorrect or unexpected runtime behavior. Vulnerabilities are security risks like SQL injection or XSS that could be exploited. Code smells are maintainability concerns that don't cause immediate errors but make the codebase harder to work with over time.

Q247**How do you configure SonarQube in Jenkins?**

Install the SonarQube Scanner plugin, configure the SonarQube server connection under "Manage Jenkins" → "Configure System", add a SonarQube analysis stage to the pipeline using the sonar-scanner command or plugin, and configure the pipeline to check and act on the quality gate result.

Q248**What are SonarQube issues, and how are they categorized?**

Issues are categorized by severity -- Blocker for issues that can cause the program to fail, Critical for significant problems likely to cause unexpected behavior, and Minor for less severe style or best-practice concerns.

Q249**How does SonarQube help manage technical debt?**

It calculates technical debt as the estimated time needed to fix all detected quality issues, which helps teams prioritize what to refactor versus what to defer, balancing that against ongoing feature development.

Q250**How does SonarQube handle multiple branches in a project?**

Branch analysis lets me analyze different branches independently and track code quality evolution per branch, ensuring new feature branches or hotfixes meet the same standard as the main branch before merging.

Q251**What is SonarLint, and how does it relate to SonarQube?**

SonarLint is an IDE plugin providing real-time code analysis as developers write code, letting issues get caught and fixed locally before a commit even happens -- complementing SonarQube by giving instant feedback rather than waiting for a CI-stage scan.

Q252**What are some best practices when using SonarQube in a CI/CD pipeline?**

Automate quality gate checks so the pipeline actually fails when the gate isn't met, aim for solid test coverage to catch untested code paths, analyze frequently -- ideally every commit or pull request -- customize quality profiles to the team's actual standards, and prioritize fixing bugs and vulnerabilities over code smells.

Q253**What is the SonarQube Scanner, and how is it used?**

The SonarQube Scanner analyzes source code and sends results to the SonarQube server, run either manually via the sonar-scanner command or as part of a CI/CD pipeline, configured through a sonar-project.properties file with the relevant project and server details.

2 2 TRIVY

Q254 What is Trivy?

Trivy is an open-source vulnerability scanner for containers and other artifacts, identifying vulnerabilities in OS packages and application dependencies across Docker images, filesystems, and Git repositories, checked against a continuously updated CVE database.

Q255 How does Trivy work?

It analyzes the image to identify its OS packages and language dependencies, checks those against its vulnerability database, and generates a report detailing findings -- severity, description, remediation guidance.

KEY POINTS TO MENTION

- Image analysis → vulnerability database check → detailed report

Q256 How can you install Trivy?

On macOS, brew install aquasecurity/trivy/trivy. Alternatively, download the release binary directly with wget and move it to /usr/local/bin, or run it as a Docker image if you'd rather not install anything locally.

Q257 How can you run a basic scan with Trivy?

trivy image scans a Docker image -- for example, trivy image nginx:latest scans the latest official nginx image for known vulnerabilities.

Q258 What types of vulnerabilities can Trivy detect?

OS package vulnerabilities across distributions like Ubuntu or Alpine, language-specific vulnerabilities in npm, Python, or Ruby dependencies, misconfigurations in infrastructure-as-code files, and known vulnerabilities in third-party libraries.

Q259 How can you integrate Trivy into a CI/CD pipeline?

Add it as a pipeline stage -- in a Jenkins declarative pipeline, a Scan stage running trivy image my-image right after the Build stage and before Deploy, failing the build if critical vulnerabilities are found.

Q260 How can you suppress specific vulnerabilities in Trivy?

A .trivyignore file lists CVE identifiers to exclude from results, one per line -- useful for documented, reviewed exceptions rather than genuinely ignoring real risk.

Q261 What are the advantages of using Trivy?

Simple to install and use, fast scanning, comprehensive coverage across multiple artifact types -- images, filesystems, Git repos -- a continuously updated vulnerability database, and easy integration into CI/CD for automated, repeatable security checks.

KEY POINTS TO MENTION

- Simplicity, speed, comprehensive coverage, continuous updates, easy CI/CD integration

Q262 Can Trivy scan local file systems and Git repositories?

Yes -- `trivy fs` scans a local directory, and `trivy repo` scans a Git repository directly, both against the same vulnerability database used for image scans.

Q263 What is the difference between Trivy and other vulnerability scanners?

Trivy stands out for ease of use with minimal setup, comprehensive coverage of both OS packages and application dependencies in one tool, fast lightweight performance suited to CI/CD, and a frequently updated vulnerability database that keeps results current.

2 3

SELENIUM

Q264 What is Selenium, and how is it used in DevOps?

Selenium is an open-source framework for automating web application testing. In DevOps I integrate it into CI/CD pipelines to automatically verify that new code changes don't break existing functionality, maintaining software quality while still enabling frequent releases.

Q265 What are the different components of Selenium?

Selenium WebDriver provides the programming interface for writing and executing test scripts; Selenium IDE is a browser extension for recording and playing back tests; Selenium Grid enables parallel test execution across machines and browsers; and Selenium RC is the older, largely deprecated component WebDriver has replaced.

Q266 How can you integrate Selenium tests into a CI/CD pipeline?

Choose a testing framework like TestNG or JUnit, write automated test scripts with Selenium WebDriver, configure the CI/CD tool to run those tests after the application is built and deployed to a test environment, and use Selenium Grid or Docker containers to run tests in parallel, isolated environments.

Q267 What challenges might you face when running Selenium tests in a CI/CD environment?

Keeping the test environment consistent with production, browser compatibility differences causing inconsistent results, flaky tests undermining trust in pipeline feedback, and resource strain from running tests in parallel if not managed carefully, leading to longer execution times.

Q268 How do you handle synchronization issues in Selenium tests?

Implicit waits set a default wait time for elements, explicit waits (WebDriverWait) wait for a specific condition before proceeding -- more flexible than implicit waits -- and fluent waits let me define polling frequency and which exceptions to ignore during the wait period.

Q269 Can you explain how you would use Selenium Grid for testing?

Start the Selenium Grid Hub as the central control point, register multiple nodes specifying available browsers and versions, point test scripts at the Grid Hub instead of a local driver, and run the tests -- the hub distributes them across nodes based on requested browser and capabilities.

Q270 How do you handle exceptions in Selenium?

Try-catch blocks around test code to catch exceptions like NoSuchElementException or TimeoutException, logging frameworks to capture error messages and stack traces, and capturing screenshots on failure with TakesScreenshot for visual evidence of what the app looked like at the time of failure.

Q271 How do you ensure the maintainability of Selenium test scripts?

The Page Object Model separates locators and page interactions from test logic, so a UI change only requires updating one page object. I also modularize tests into reusable methods, use consistent naming conventions, and keep everything in version control to track changes and collaborate.

Q272 How can you run Selenium tests in headless mode?

Configure the browser driver's options for headless execution -- for Chrome, adding the --headless argument to ChromeOptions before initializing the driver -- so tests run without a GUI, which is exactly what's needed in a CI/CD environment.

Q273 What is the role of Selenium in the testing pyramid?

Selenium sits in the UI testing layer, handling end-to-end validation of user interactions, and should complement -- not replace -- unit tests at the base and integration tests in the middle. Using it wisely within that pyramid, rather than over-relying on it, optimizes both coverage and test-suite speed.

Q274 What is Nexus Repository Manager?

Nexus Repository Manager is a repository management tool for storing and sharing software artifacts, supporting formats like Maven, npm, NuGet, and Docker. It centralizes binary management, improving dependency management and CI/CD integration across teams.

Q275 What are the main features of Nexus Repository Manager?

Support for multiple repository formats, proxying and caching remote repositories to speed up builds, straightforward artifact upload/storage/retrieval, fine-grained security and access control, seamless CI/CD tool integration, and repository health checks for monitoring performance.

Q276 How do you configure Nexus Repository Manager?

Install Nexus, access the web interface, create the repositories needed -- hosted, proxy, or group -- configure security roles and permissions, set up proxy repository remote URLs and caching if needed, and point build tools like Maven or npm at the Nexus repository for dependency resolution.

Q277 What is the difference between a hosted, a proxy, and a group repository in Nexus?

A hosted repository stores your own uploaded artifacts, typically for internal projects. A proxy repository caches artifacts from a remote source like Maven Central, speeding up builds and reducing internet dependency. A group repository aggregates multiple hosted and proxy repositories behind a single URL, simplifying dependency resolution for consumers.

KEY POINTS TO MENTION

- Hosted: your own artifacts
- Proxy: cached external dependencies
- Group: aggregates repos behind one endpoint

Q278 How do you integrate Nexus Repository Manager with Jenkins?

Install the Nexus Artifact Uploader plugin, configure Nexus repository settings within the Jenkins job, publish artifacts to Nexus via post-build actions after a successful build, and update build tools like Maven in Jenkins to resolve dependencies from the Nexus repository.

Q279**What are the security features in Nexus Repository Manager?**

User authentication support for LDAP and Crowd, role-based access control for who can access or modify repositories, SSL support for secure communication, and audit logs tracking user actions for security and compliance purposes.

Q280**How can you monitor the health and performance of Nexus Repository Manager?**

The Nexus web UI provides basic usage and performance stats, built-in health check reports monitor repository status, and integrating Nexus with external tools like Prometheus or Grafana gives more detailed metrics and alerting on performance and usage.

2 5

COMBINED: GITHUB ACTIONS, ARGOCD & KUBERNETES

Q281

How would you deploy a Kubernetes application using GitHub Actions and ArgoCD?

GitHub Actions handles the CI side -- linting manifests, running tests, building and pushing the image, then updating the Kubernetes manifests or Helm values in the Git repo ArgoCD watches. ArgoCD picks up that change and automatically syncs the cluster to match, so GitHub Actions never needs direct write access to the cluster itself.

Q282

Can you explain the GitOps workflow in Kubernetes using ArgoCD and GitHub Actions?

Developers push code or manifest changes to Git, a GitHub Actions workflow validates those changes and updates the Kubernetes manifests accordingly, and ArgoCD continuously monitors the repository, automatically syncing the live cluster to match the desired state declared in Git.

Q283

How do you manage secrets for Kubernetes deployments in GitOps using GitHub Actions and ArgoCD?

GitHub Actions secrets handle anything the CI workflow itself needs, while in the cluster I use Sealed Secrets or HashiCorp Vault, or native Kubernetes Secret management, so actual secret values are never committed as plaintext into the Git repository ArgoCD is watching.

26

GITLAB

Q284 What is GitLab?

GitLab is a web-based DevOps lifecycle tool providing Git repository management alongside CI/CD, issue tracking, and monitoring, integrating multiple stages of the software development lifecycle into a single application.

Q285 How does GitLab CI/CD work?

The pipeline is defined in a `.gitlab-ci.yml` file at the repo root, specifying stages, jobs, and scripts. GitLab Runner picks up that configuration and executes the jobs on the configured runner -- shared, group, or project-specific.

Q286 What is a GitLab Runner?

A GitLab Runner is the application that processes CI/CD jobs, installable on various platforms and capable of running jobs in different environments like Docker or shell, either shared across projects or dedicated to a specific one.

Q287 What is the difference between GitLab and GitHub?

GitLab bundles integrated CI/CD, issue tracking, and project management natively in one platform, making it well suited for full DevOps workflows, and offers strong self-hosting support. GitHub is more focused on social/open-source coding, with GitHub Actions adding CI/CD capability more recently, and primarily operates as a cloud service.

Q288 Can you explain the GitLab branching strategy?

A common approach is Git Flow -- Master/Main as the stable production version, Develop as the integration branch, feature branches created from Develop for specific work, release branches for preparing production releases, and hotfix branches for urgent fixes directly off Master.

Q289 What is the purpose of a `.gitlab-ci.yml` file?

It defines the CI/CD pipeline configuration for a project -- stages, jobs, scripts, and the conditions under which jobs run -- and is essential for automating build, test, and deployment in GitLab CI/CD.

Q290 How do you handle merge conflicts in GitLab?

Merge the conflicting branch into the current branch locally, use `git merge` or `git rebase` to resolve conflicts in an editor, commit the resolved changes, and push back to the repository -- or resolve conflicts directly through GitLab's web interface on the merge request.

Q291**What are GitLab CI/CD pipelines?**

Pipelines are the automated processes defined in `.gitlab-ci.yml` that build, test, and deploy code -- made up of stages that run sequentially, each containing jobs that run concurrently, ensuring consistent delivery and automating repetitive tasks.

Q292**What is the purpose of GitLab Issues?**

GitLab Issues track tasks, bugs, and feature requests within a project, letting teams create, assign, comment on, and close them, with labels, milestones, and due dates helping prioritize and organize work.

Q293**Explain the concept of tags in GitLab.**

Tags reference specific points in a repository's history, typically marking release versions or milestones. They're immutable snapshots of a particular commit, either annotated with extra information or lightweight, and are useful for managing releases and deployments.

Q294 What is Python's role in DevOps?

Python's simplicity and rich ecosystem make it useful for automating infrastructure as code alongside tools like Terraform and Ansible, scripting steps in CI/CD pipelines, and building monitoring and logging integrations with tools like Prometheus or Grafana APIs where the built-in tooling doesn't cover a specific need.

Q295 How can you use Python in Jenkins pipelines?

I call Python scripts directly within a pipeline stage using the sh step -- for example sh 'python3 script.py' inside a stage block -- to automate testing, packaging, or deployment steps as part of the overall Jenkins pipeline.

Q296 How would you manage environment variables in Python for a DevOps project?

The os module reads environment variables directly -- `os.getenv("DATABASE_URL", "default_value")` -- and for more structured or sensitive management I'd use something like python-dotenv locally or Docker secrets in containerized deployments.

Q297 How do you use Python to interact with a Kubernetes cluster?

The official kubernetes Python client lets me load the kubeconfig and call the API directly -- listing pods in a namespace with `v1.list_namespaced_pod()`, for example -- which is also how I've built custom automation like scripts that find and report every pod stuck in a bad state across a cluster.

Q298 How do you use Python to monitor server health in DevOps?

The psutil library gives direct access to CPU and memory usage -- `psutil.cpu_percent()` and `psutil.virtual_memory()` -- which I can extend to push those metrics into Prometheus or Grafana for ongoing monitoring rather than just a one-off check.

Q299 What is the use of the subprocess module in DevOps scripting?

subprocess lets a Python script spawn and manage other processes, capturing their output and return codes, which is useful for automating shell commands, deploying code, or wrapping CLI tools like Docker directly inside a Python automation script.

Q300**How do you handle exceptions in Python scripts for DevOps automation?**

try-except blocks around anything that can fail, catching specific exceptions like `subprocess.CalledProcessError` rather than a bare except, logging the actual error with useful context, and optionally triggering a retry mechanism for genuinely transient failures.

Q301**Can you explain how Python works with cloud services in DevOps?**

Cloud SDKs like Boto3 for AWS let Python interact directly with services -- listing S3 buckets, managing EC2 instances -- which is how I've automated infrastructure provisioning, deployment, and scaling tasks beyond what the CLI conveniently covers.

Q302**How do you use Python for log monitoring in DevOps?**

Reading and filtering log files directly in Python, or integrating with a stack like ELK for larger scale, lets me search for patterns like "ERROR" and trigger alerts through Slack or email notifications when specific patterns show up.

Q303**How would you use Python in a Dockerized DevOps environment?**

Python often is the application logic running inside containers, and separately, the Docker SDK for Python lets me manage containers programmatically -- pulling images and running containers directly from a script -- which I've used to automate deployment or orchestration tasks beyond plain docker CLI commands.

Closing Note

This is the exhaustive edition — all 303 questions across every topic in the source question bank, from networking and subnetting fundamentals through AWS, Azure, Git, CI/CD, Docker, Kubernetes internals, Terraform, Ansible, monitoring, security, Jenkins, Linux, SonarQube, Trivy, Selenium, Nexus, GitLab, and Python for DevOps. Every answer is written the way I'd actually say it out loud in an interview, so treat this as a full rehearsal script: adapt the specifics to your own project experience before you walk in.

Compiled and elaborated by Arvind Verma.